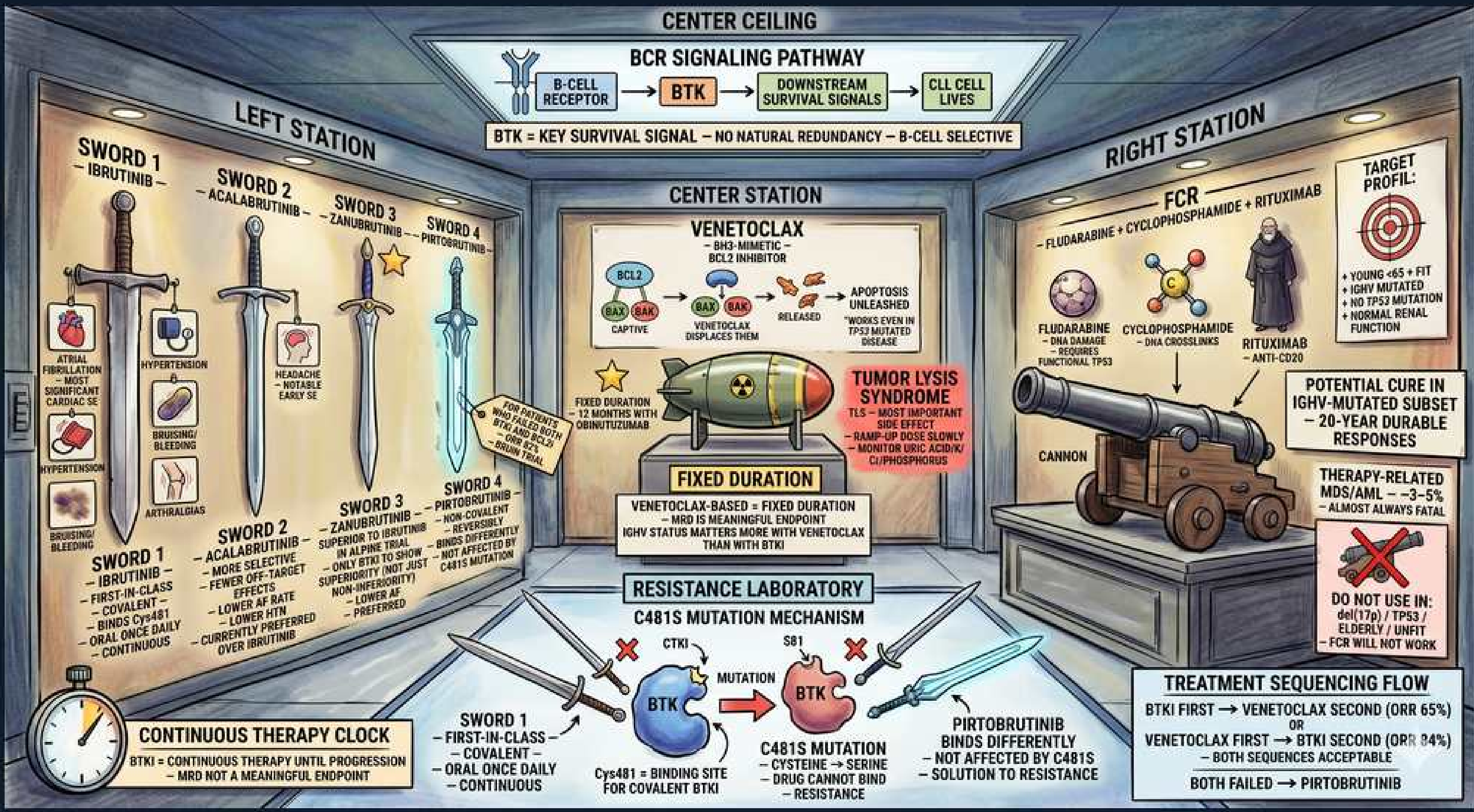
CLL Targeted Therapy: BTK Inhibitors, Venetoclax, FCR & Resistance



1. BCR -> BTK -> survival signaling

WHAT YOU SEE

Center ceiling: B-cell receptor -> BTK -> CLL cell lives

WHAT IT ENCODES

B-cell receptor (BCR) signaling through Bruton tyrosine kinase (BTK) is the key survival pathway in CLL. BTK is B-cell selective with no natural redundancy, making it an ideal drug target.

BOARD TRAP

BTK blockade works because BCR signaling is non-redundant for CLL B-cell survival — this is why BTK inhibitors are broadly effective, including in del(11q) and unmutated IGHV.

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2. Ibrutinib (1st-gen BTKi)

WHAT YOU SEE

Sword 1: ibrutinib, continuous oral therapy

WHAT IT ENCODES

Ibrutinib = first-in-class covalent BTK inhibitor, continuous oral therapy. Effective regardless of IGHV/FISH but has cardiac and bleeding toxicity.

BOARD TRAP

Ibrutinib's main toxicities are ATRIAL FIBRILLATION, hypertension, and bleeding — these (not myelosuppression) drive switching to second-generation agents.

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3. Acalabrutinib / zanubrutinib

WHAT YOU SEE

Swords 2 and 3: acalabrutinib, zanubrutinib, fewer side effects

WHAT IT ENCODES

Second-generation covalent BTKi (acalabrutinib, zanubrutinib) are more selective with LESS atrial fibrillation and bleeding than ibrutinib, with equal/better efficacy.

BOARD TRAP

Switch ibrutinib -> acala/zanu for cardiac toxicity (afib), not for resistance — second-gen still bind the same C481 site and don't overcome C481S resistance.

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4. Pirtobrutinib (non-covalent)

WHAT YOU SEE

Sword 4: pirtobrutinib, reversible binding

WHAT IT ENCODES

Pirtobrutinib is a non-covalent (reversible) BTK inhibitor that binds independently of C481, so it retains activity after covalent BTKi failure due to C481S mutation.

BOARD TRAP

Pirtobrutinib is the answer for C481S-mutated, covalent-BTKi-resistant CLL — covalent agents (ibrutinib/acala/zanu) all fail at that mutation.

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5. Venetoclax = BCL2 inhibitor

WHAT YOU SEE

Center station: venetoclax bomb releasing apoptosis, BH3-mimetic

WHAT IT ENCODES

Venetoclax is a BH3-mimetic that inhibits anti-apoptotic BCL2, displacing pro-apoptotic proteins to trigger apoptosis. Used as fixed-duration therapy, often with obinutuzumab.

BOARD TRAP

Venetoclax restores apoptosis via BCL2 inhibition — it works even in del(17p)/TP53 because its mechanism is downstream and p53-independent, unlike chemoimmunotherapy.

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6. Tumor lysis syndrome — ramp-up

WHAT YOU SEE

Radioactive bomb labeled tumor lysis syndrome with weekly dose ramp-up

WHAT IT ENCODES

Venetoclax causes TUMOR LYSIS SYNDROME, mitigated by a 5-week dose ramp-up, hydration, allopurinol/rasburicase, and lab monitoring. Risk stratified by tumor burden/renal function.

BOARD TRAP

The signature venetoclax toxicity is TLS, managed by gradual dose ramp-up — abrupt full dosing can cause fatal hyperkalemia/AKI.

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7. Venetoclax fixed duration

WHAT YOU SEE

Sign: fixed duration, MRD-driven, time-limited

WHAT IT ENCODES

Venetoclax-based therapy is given for a fixed/time-limited duration (often MRD-guided), unlike continuous BTKi — a key practical distinction for patients wanting a defined treatment course.

BOARD TRAP

BTK inhibitors are CONTINUOUS until progression; venetoclax-based regimens are FIXED-DURATION — don't conflate the two treatment philosophies.

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8. FCR (chemoimmunotherapy)

WHAT YOU SEE

Right station: fludarabine + cyclophosphamide + rituximab cannon

WHAT IT ENCODES

FCR (fludarabine, cyclophosphamide, rituximab) is older chemoimmunotherapy. Durable in YOUNG, fit, IGHV-MUTATED patients without del(17p)/TP53; ~20-year durable responses in that subset.

BOARD TRAP

FCR is contraindicated in del(17p)/TP53 and unmutated IGHV; it also carries therapy-related MDS/AML risk — modern targeted agents have largely replaced it.

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9. FCR ideal target profile

WHAT YOU SEE

Target profile: young, fit, IGHV-mutated, no del(17p), normal renal

WHAT IT ENCODES

The narrow FCR niche: young, fit, IGHV-mutated, del(13q)/normal cytogenetics, good renal function — the only group with potential functional cure from chemoimmunotherapy.

BOARD TRAP

FCR's durable benefit is restricted to IGHV-MUTATED without TP53 loss; giving it to unmutated or del(17p) patients yields poor, short responses.

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10. C481S resistance mutation

WHAT YOU SEE

Resistance laboratory: C481S mutation at the covalent BTK binding site

WHAT IT ENCODES

C481S substitutes the cysteine that covalent BTKi bind, abolishing irreversible inhibition -> resistance to ibrutinib/acalabrutinib/zanubrutinib. Pirtobrutinib (non-covalent) overcomes it.

BOARD TRAP

C481S = resistance to COVALENT BTKi only; the solution is the non-covalent pirtobrutinib — switching between covalent agents does NOT help.

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11. Venetoclax resistance (BCL2/BTG1)

WHAT YOU SEE

Resistance station: BCL2 G101V binding-site mutation reduces venetoclax binding

WHAT IT ENCODES

Acquired BCL2 mutations (e.g. G101V) reduce venetoclax binding affinity, driving relapse after BCL2-targeted therapy. Mechanistically distinct from BTKi C481S resistance.

BOARD TRAP

Venetoclax resistance arises from BCL2 binding-site mutations, NOT C481S — the two targeted classes fail by different mechanisms, which is why sequencing both still works.

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12. Treatment sequencing flow

WHAT YOU SEE

Bottom: BTKi first -> venetoclax second; venetoclax first -> BTKi second; both failed -> pirtobrutinib

WHAT IT ENCODES

Sequencing: start BTKi or venetoclax (either order, both effective). On failure, switch to the other class. After BOTH covalent BTKi and venetoclax fail -> pirtobrutinib, then CAR-T/allo-SCT.

BOARD TRAP

After failing both a covalent BTKi and venetoclax, the next targeted option is pirtobrutinib — not re-trying a covalent BTKi or restarting chemoimmunotherapy.
